

CO3-AT2-MCQ

Google Forms Score / Response

Total Score: 26/32

19. A pipeline hazard is a situation that:

Answer: B) Prevents the next instruction from executing normally **Status:** Correct

30. Intel's Hyper-Threading technology is an implementation of:

Answer: B) Simultaneous Multithreading (SMT) **Status:** Correct

25. Out-of-order execution allows a processor to:

Answer: B) Execute independent instructions before earlier stalled instructions **Status:** Correct

5. The operation $R3 \leftarrow R1 + R2$ performs:

Answer: Arithmetic operation **Status:** Correct

18. Ideally, after a pipeline is filled, one instruction can be completed:

Answer: A) Every clock cycle **Status:** Correct

24. A superscalar processor is capable of:

Answer: B) Issuing multiple instructions in a single clock cycle **Status:** Correct

7. The XOR operation produces a 1 when:

Answer: Inputs are different **Status:** Correct

15. In a typical five-stage instruction pipeline, which stage is responsible for performing arithmetic or logical operations?

Answer: EX **Status:** Correct

17. Pipeline performance is primarily measured using:

Answer: A) Instruction throughput and execution time **Status:** Correct

9. In a three-bus organization, the ALU can typically receive operands from:

Answer: Two separate buses **Status:** Correct

Registration Number

Response: 192524275 **Status:** Incorrect

26. The primary benefit of out-of-order execution is:

Answer: A) Better utilization of processor resources **Status:** Correct

10. Multiple-bus organization improves processor performance mainly by:

Answer: Allowing simultaneous movement of data **Status:** Correct

27. Speculative execution refers to:

Answer: A) Executing instructions based on predicted future control flow **Status:** Correct

3. Which operation is an arithmetic micro-operation?

Answer: Addition **Status:** Correct

6. Which operation is generally used to set selected bits of a register to 1?

Answer: OR **Status:** Correct

20. Which of the following is a data hazard?

Answer: B) RAW dependency **Status:** Correct

12. Consider the operation $R3 \leftarrow R1 + R2$. In a two-input-bus organization, R1 and R2 can be placed on:

Answer: Two different input buses **Status:** Correct

28. The dispatch unit in a superscalar processor is responsible for:

Answer: B) Sending instructions to appropriate execution units **Status:** Correct

14. Which of the following is a typical stage of an instruction pipeline?

Answer: Instruction Fetch **Status:** Correct

29. Simultaneous Multithreading (SMT) allows a processor core to:

Answer: A) Execute instructions from multiple threads concurrently **Status:** Correct

8. The primary advantage of a multiple-bus organization is:

Answer: Increased parallel data transfers **Status:** Correct

23. A structural hazard occurs when:

Correct answer: A) Two instructions require the same hardware resource simultaneously **Status:** Incorrect

16. The correct sequence of a typical five-stage pipeline is:

Correct answer: IF → ID → EX → MEM → WB **Status:** Incorrect

Name of the Student

Response: Yashwanth.Gudi **Status:** Incorrect

1. Which notation is commonly used to represent a transfer of data from register R1 to register R2?

Answer: $R1 \rightarrow R2$ **Status:** Correct

13. The main purpose of instruction pipelining is to:

Answer: Increase instruction throughput **Status:** Correct

11. In a multiple-bus system, buses are primarily used to:

Answer: Transfer data between processor components **Status:** Correct

4. Which of the following is a logical micro-operation?

Correct answer: AND **Status:** Incorrect

22. Which hazard is directly associated with branch instructions?

Answer: A) Control hazard **Status:** Correct

2. In register transfer language, the statement $R2 \leftarrow R1$ indicates that:

Correct answer: Contents of R1 are copied into R2 **Status:** Incorrect

21. A Read After Write (RAW) hazard occurs when:

Answer: A) An instruction reads a value before a previous instruction writes it **Status:** Correct